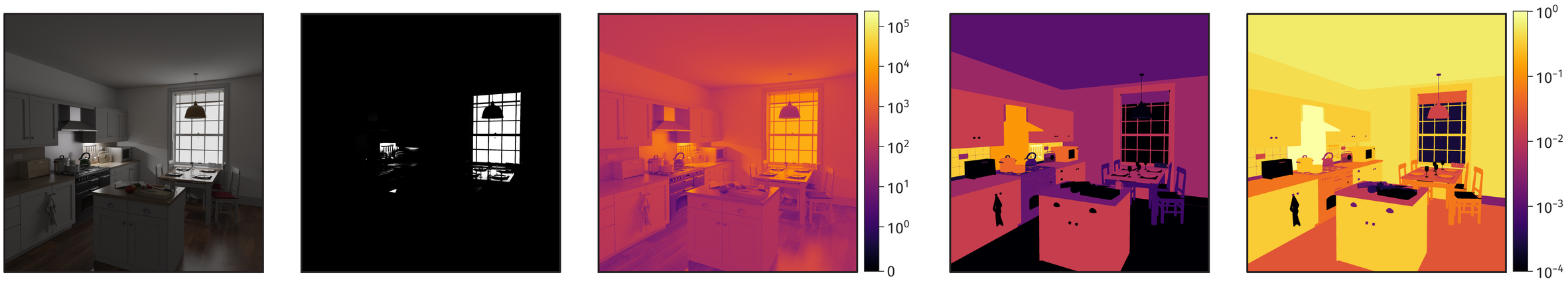
Scene Parameter Saliency via Differentiable Light Transport

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Introduction

Gradients from a differentiable renderer are *not* just for optimisation. For a value measured on the image, the renderer can compute the gradient with respect to every scene parameter, such as material colours, roughness and light strength. These gradients are normally only used to optimise the scene. Instead, we directly visualise the gradients as saliency maps for deeper inspection. Grad-CAM [1] uses gradients to show which pixels matter for a classification. We use them to show which scene parameters matter for a metric. The gradient includes light that bounces between surfaces, so it can find causes that are hard to see in the image.

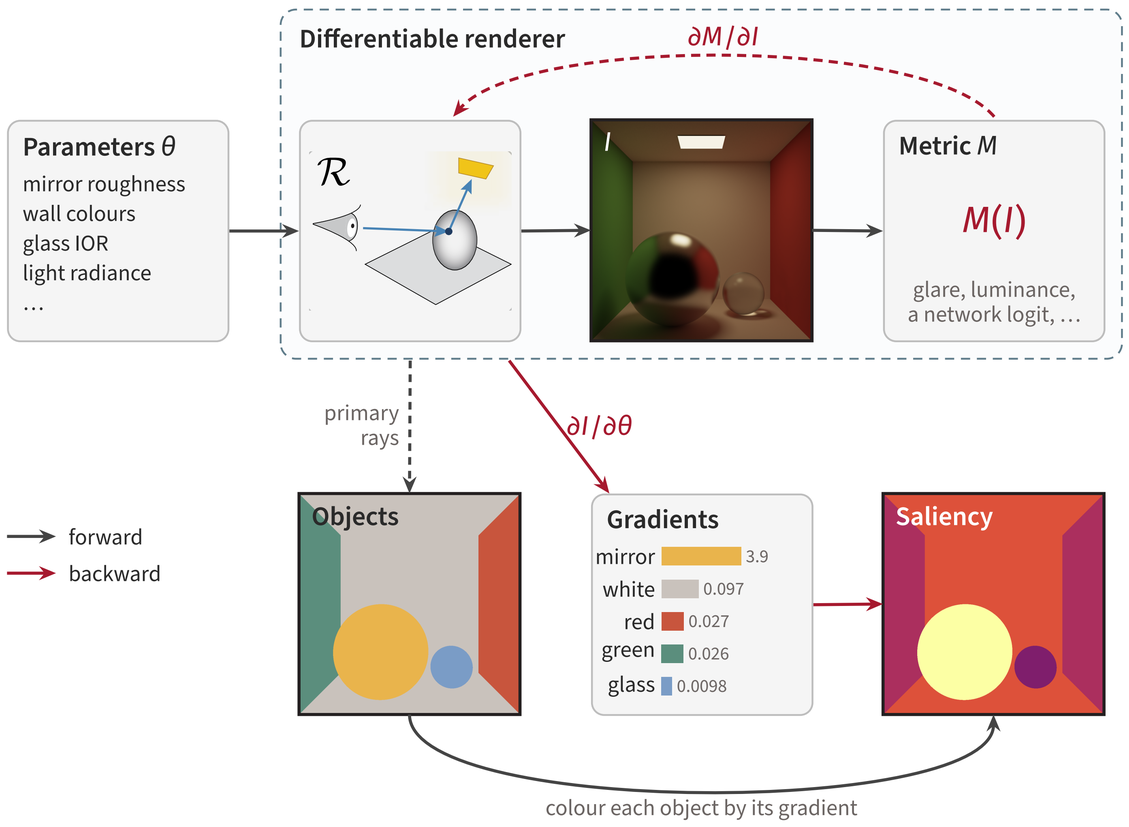


(a) Beauty render (b) Glare mask (c) Luminance (d) **Glare saliency** (e) **Mean-luminance saliency**

Example: discomfort glare in a kitchen. The glare mask (b) and luminance map (c) show where the glare is, but not what causes it. The glare saliency map (d) colours each surface by how much its material affects a glare metric, and shows that most of the glare comes from the **metal utensils**. For the same scene, the mean-luminance map (e) highlights the **extractor hood, ceiling and walls** instead, because they are large and bright. Different metrics give different saliency maps. Saliency is scaled on a log scale so the largest value is 1.

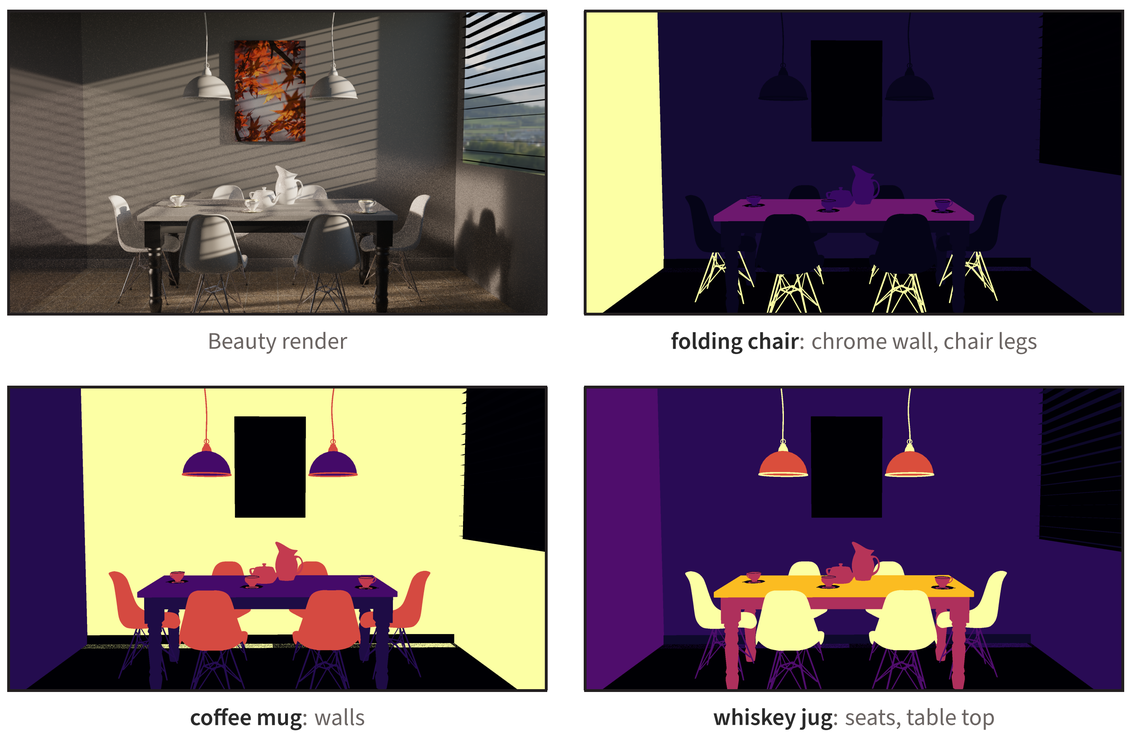
Method

One backward pass gives the saliency of every parameter, which we then map onto the scene. We render the image $I = R(\theta)$, compute a metric $M(I)$, and backpropagate through both to get the gradient $g_i = \partial M / \partial \theta_i$. It tells us how much the metric changes when θ_i changes, including light reflected from other surfaces. We then find which object is visible at each pixel and colour it by the size of its gradient, $\|g_i\|$. The size shows how important a parameter is and the sign shows which way to change it. For parameters with several values, such as an RGB colour, we use the length of the gradient vector. In the Cornell box below, the metric is glare and the mirror has the largest gradient. We use the Mitsuba 3 renderer [3].



Neural networks

The metric only needs to be differentiable, so it can also be a neural network. We backpropagate the ResNet-50 [2] score for one class through the network and then through the renderer, so the map shows the sensitivity of both together. **Each class gives a different saliency map.** These maps use a linear scale, so the most salient material stands out.



Beauty render

folding chair: chrome wall, chair legs

coffee mug: walls

whiskey jug: seats, table top

The left wall is a chrome mirror that reflects the chairs. For the folding chair class the wall has high saliency, which means the network also uses the **reflected chairs**.

In principle, the method applies to any objective function that is differentiable with respect to the rendered image, such as perceptual similarity scores or other learned models.

Applications

- **Lighting design:** find what causes glare with one backward pass. A parametric sweep needs a separate simulation for each parameter.
- **Other design metrics:** daylight and visual comfort measures can be used in the same way.
- **Vision models:** see which materials in a scene a network responds to.
- **Optimisation:** use the map to choose which parameters to optimise and which to keep fixed.
- **Other renderers:** the method works with any differentiable renderer.

Limitations & next steps

- The gradient only describes small changes and ignores interactions between parameters. Very small gradients are affected by render noise.
- Metrics with hard thresholds need a differentiable approximation.
- Each object currently gets a single value. Textures and spatially varying materials are left to future work.
- Next: signed maps, interactive editing, and second-order analysis.

[1] R. R. Selvaraju et al. Grad-CAM. ICCV 2017. [2] K. He et al. Deep residual learning for image recognition. CVPR 2016. [3] W. Jakob et al. Mitsuba 3 renderer. 2022.

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Project page: linasberesna.com/views/bmvc2026.html
Code: github.com/LinasBeres/dr-scene-parameter-saliency